AttendanceOps

Product Management Case Study

Django-Based Employee Attendance and Payroll Analytics Dashboard

1. Product Overview

Product: AttendanceOps

What it does: AttendanceOps is a Django-based web application that processes employee check-in and check-out records, calculates working hours and estimated payroll, detects unusual attendance patterns, and provides an interactive dashboard with an AI-powered chatbot for answering attendance-related questions.

Target users: The product is designed for manufacturing companies in India, primarily targeting HR teams, payroll administrators, and factory managers responsible for tracking employee attendance and calculating wages.

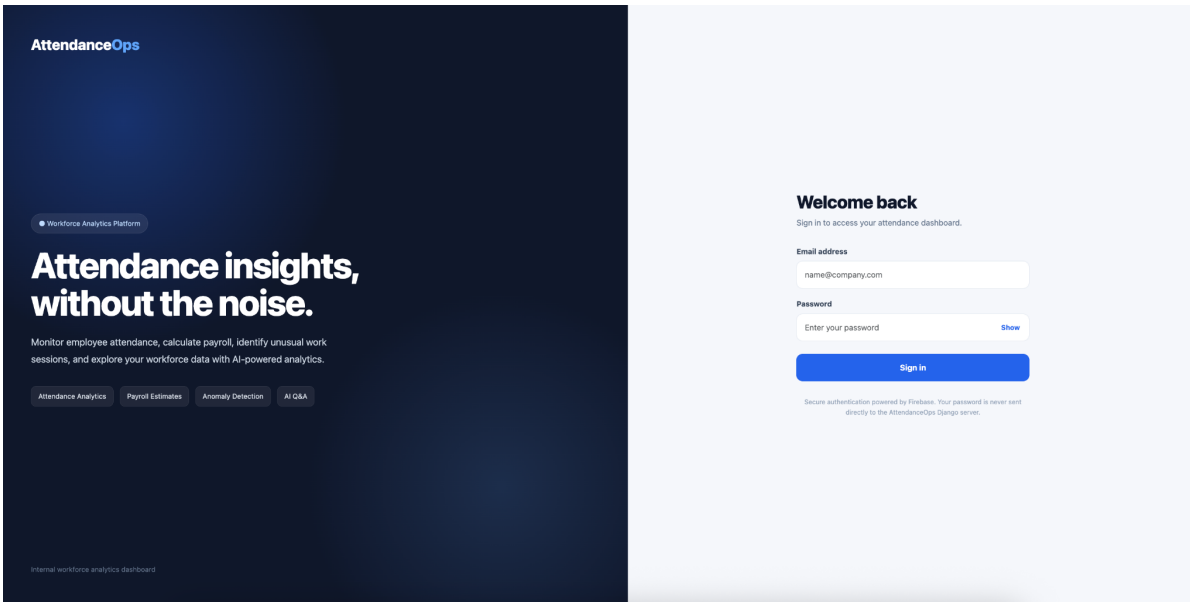


Figure 1: AttendanceOps secure login experience.

2. User Problem

Primary users: Factory managers, HR personnel, and payroll administrators responsible for tracking employee working hours and preparing payroll.

Main problem: Attendance data often consists of individual check-in and check-out records that are difficult to interpret. Managers must manually match these records, calculate total working hours, and identify irregularities, making payroll preparation time-consuming and prone to errors.

Why this matters: Incorrect attendance calculations can result in employees being underpaid or overpaid. Manual processing also consumes administrative time and makes payroll disputes more difficult to resolve.

Current workaround: Managers manually examine attendance logs, match check-in and check-out timestamps, calculate daily or monthly working hours using spreadsheets, and investigate missing or unusual records.

3. Product Goal

Reduce the time managers spend preparing attendance-based payroll reports by **50%**, while minimizing calculation errors and making attendance anomalies easier to identify.

This is a proposed product target that would require validation through real user testing.

4. User Needs

The three primary user needs are:

1. **Accurate attendance tracking:** Managers need to automatically calculate working hours from employee check-in and check-out records without manually matching timestamps.

2. **Reliable payroll estimation:** Managers need to calculate estimated wages based on hours worked and identify anomalies or discrepancies before finalizing employee payroll.

3. **Accessible attendance insights:** Managers need to search, filter, and understand employee attendance data without manually reviewing large Excel datasets.

5. Possible Features

- **Automated attendance processing:** Match employee check-in and check-out records to calculate total working hours.
- **Payroll estimation:** Calculate estimated employee wages based on total hours worked and a specified hourly rate.
- **Attendance anomaly detection:** Flag unusually long or short shifts, missing records, and abnormal attendance patterns.
- **Reporting dashboard:** Display employee-level and daily summaries, charts, and attendance reports with CSV and Excel export capabilities.
- **AI-powered chatbot:** Allow managers to ask natural-language questions about attendance data.

6. Feature Prioritization

Priority 1: Automated Attendance Processing

Why it matters: Accurate working-hour calculations form the foundation of the application. Without correctly matching employee check-in and check-out records, payroll calculations and reports cannot be trusted.

Why now: This is the core functionality that must exist before other features can deliver reliable value.

Priority 2: Payroll Estimation and Summary Dashboard

Why it matters: Managers need a clear overview of employee working hours and estimated wages to prepare payroll efficiently and reduce manual calculations.

Why now: Once attendance records are processed accurately, displaying working hours and estimated wages provides immediate operational value to managers.

Priority 3: Attendance Anomaly Detection

Why it matters: Incorrect or unusual attendance records can lead to inaccurate payroll estimates. Automatically identifying suspicious sessions helps managers investigate potential payroll issues before finalizing payments.

Why now: Anomaly detection improves the reliability of the attendance workflow and reduces the risk of overlooking discrepancies.

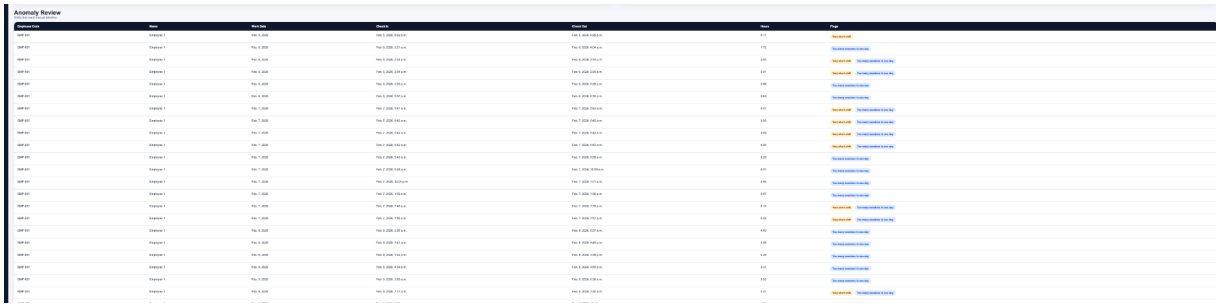


Figure 2: Anomaly Review surfaces unusual attendance sessions for manager review.

7. What I Would Not Build Yet

Feature to defer: Further development of the Ollama-powered AI chatbot.

Reasoning: Accurate attendance calculations, reliable payroll estimation, and anomaly detection should take priority because they directly address the users' main problems.

Although the chatbot improves accessibility through natural-language queries, its additional functionality should be prioritized only after the core attendance and payroll workflows are reliable and validated.



Figure 3: Attendance Q&A allows managers to ask natural-language questions about structured attendance data.

8. Product Tradeoffs

Speed vs. Quality

Accuracy should be prioritized over speed for attendance and payroll calculations because incorrect working hours can directly affect employee compensation. The initial implementation should remain simple while maintaining reliable calculations.

More Features vs. Simpler MVP

I would focus on a simpler MVP consisting of automated attendance processing, payroll estimation, and basic reporting rather than immediately introducing advanced features.

This allows the product to validate whether it solves the primary user problem before increasing complexity.

User Convenience vs. Engineering Complexity

An AI chatbot makes it easier for managers to retrieve information using natural language, but it also introduces model integration complexity, response latency, and the possibility of inaccurate answers.

I would prioritize structured dashboards and reports before expanding chatbot capabilities.

Automation vs. User Control

The application should automatically calculate working hours and flag attendance anomalies while allowing managers to review discrepancies and approve finalized payroll amounts.

Automated payroll estimates should not directly trigger employee payments without appropriate review and approval.

9. Success Metrics

Primary Metric: Payroll Preparation Time

Measure the average time required for a payroll administrator to prepare a monthly attendance-based payroll report.

Target: Reduce preparation time by 50% compared with the existing manual workflow.

Secondary Metric: Attendance Processing Automation Rate

Measure the percentage of eligible attendance records successfully converted into valid work sessions without manual intervention.

Target: Achieve at least a 95% successful processing rate for records with valid matching punches.

Quality / Guardrail Metric: Working-Hour Calculation Accuracy

Measure the percentage of validated work sessions whose calculated working hours match manually verified expected results within a defined tolerance.

Target: Achieve at least 99% accuracy on a representative validated dataset.

Note: These are proposed product targets rather than measured outcomes. Actual targets would require user testing and data validation.

10. MVP User Flow

1. **User action:** A factory manager opens the attendance dashboard, selects a date range, and enters the applicable hourly pay rate.

2. **System action:** The application retrieves attendance records, matches check-in and check-out timestamps, calculates working hours and estimated payroll, and flags unusual attendance sessions.

3. **User sees:** The dashboard displays total working hours, estimated wages, employee-level summaries, and flagged attendance anomalies.

4. **Next action:** The manager reviews discrepancies, inspects individual employee records, and exports attendance reports to CSV or Excel for further payroll processing.

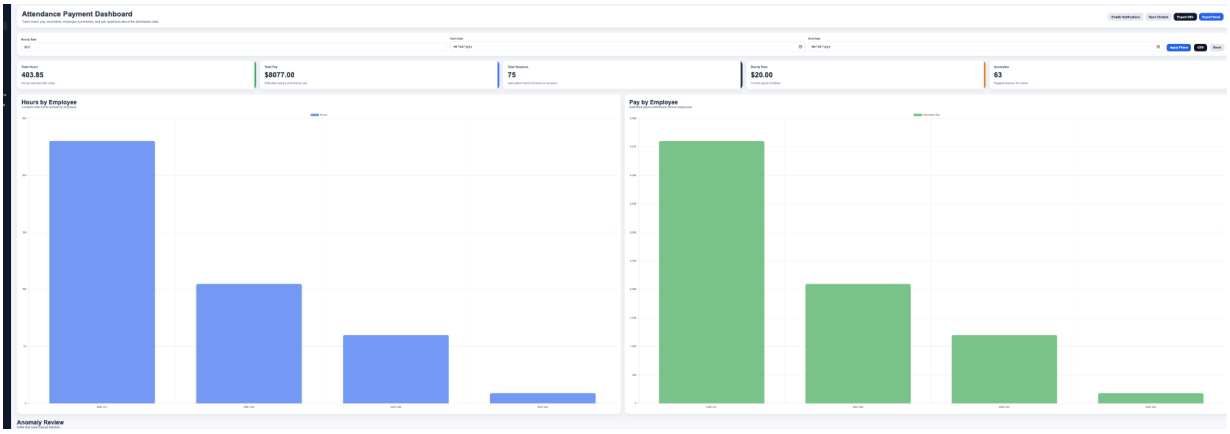


Figure 4: AttendanceOps dashboard showing attendance metrics, payroll estimates, employee-level analytics, and anomaly counts.

11. Risks, Assumptions, and User Validation

Biggest Risk

The primary risk is inaccurate or incomplete attendance records. Missing check-ins, missing check-outs, or incorrect timestamps could result in inaccurate working-hour calculations and payroll estimates, reducing managers' trust in the product.

Another important risk involves security and ensuring that only authorized users can access or modify sensitive employee and payroll information.

Key Assumptions

I assume that manufacturing managers spend significant time manually processing attendance records and that automated calculations and reporting would meaningfully reduce their workload.

I also assume that different users require different levels of access to attendance and payroll information, making role-based permissions important for operational use.

These assumptions would need to be validated with actual users.

What I Would Learn From Users

I would interview factory managers, HR personnel, and payroll administrators to understand how attendance-based payroll is currently processed.

Specifically, I would investigate:

- Which attendance-processing tasks consume the most time.
- How frequently missing punches and payroll discrepancies occur.
- Which users need access to attendance records and payroll information.
- What permissions different roles require.
- Whether managers trust automated calculations and what review processes they expect.

I would also compare manual and automated payroll preparation workflows to evaluate whether the product reduces administrative effort.

How I Would Validate the MVP

Before expanding the product, I would conduct a small pilot with 3–5 factory managers, HR personnel, and payroll administrators to validate the core assumptions.

1. **Understand the existing workflow:** Interview users to understand how they currently process attendance records, calculate payroll, and resolve discrepancies.

2. **Test the core functionality:** Ask participants to complete attendance-processing and payroll-reporting tasks using AttendanceOps with representative sample data.

3. **Compare performance:** Measure the time required to prepare attendance-based payroll reports manually versus using AttendanceOps. Compare calculated working hours against manually verified records.

4. **Gather feedback:** Identify usability issues, missing functionality, and situations where users do not trust automated calculations.

Decision Criteria: I would prioritize improving calculation accuracy and resolving workflow issues before expanding AI chatbot capabilities.

The MVP would be considered promising if users complete core payroll preparation tasks more efficiently without introducing unacceptable calculation errors.

12. Next Version

Proposed Features: Attendance correction workflows and improved role-based access control (RBAC).

Why: The application identifies unusual attendance sessions, but managers also need a structured way to investigate and resolve discrepancies.

A correction and approval workflow would allow authorized managers to review missing punches, correct attendance records, and approve finalized working hours before payroll processing.

Basic authentication and appropriate access restrictions should be treated as essential security requirements. More granular role-specific permissions and advanced approval workflows can be developed further in subsequent iterations.

These improvements would make the application more reliable and suitable for operational use.

13. 30-Second Product Pitch

Manufacturing companies often struggle to track employee attendance and calculate payroll because their data consists of individual check-in and check-out records.

To address this problem, I developed **AttendanceOps**, a Django-based dashboard that automatically converts attendance records into working hours, estimates payroll, and flags unusual attendance patterns.

Managers can view employee-level summaries, export CSV and Excel reports, and ask attendance-related questions through an AI-powered chatbot. The application also uses Firebase Admin session cookies to support authenticated sessions.

The product aims to reduce manual payroll preparation, improve calculation accuracy, and help ensure employees are compensated fairly for the hours they work.